

AMARNATH ALLAMRAJU

Indianapolis, Indiana | 317-552-3754 | amarnathallamraju@gmail.com | [LinkedIn](#) | [GitHub](#) | [SQLit](#)

PROFESSIONAL SUMMARY

Data Engineer with expertise in designing and optimizing scalable ETL/ELT pipelines using Delta Lake, PySpark, and Airflow across cloud platforms including Azure, AWS, and BigQuery. Skilled in Snowflake and relational databases with strong SQL and Python capabilities. Developed medallion architecture and data governance policies to reduce onboarding time by 35% and achieve 99% SLA adherence, accelerating analytics delivery for enterprise users.

TECHNICAL SKILLS

- **Data Warehouse & Platforms:** Snowflake (SQL/T-SQL, query optimization, warehouse tuning, data modeling), Microsoft Azure (Data Lake, Synapse), Medallion Architecture (Bronze/Silver/Gold), Data Warehouse Design & Optimization, ETL/ELT Pipeline Architecture, Schema Design & Performance Tuning, Relational Databases
- **Cloud & Storage:** Azure Data Lake & Synapse, AWS (S3, Glue, Lambda, Redshift), GCP BigQuery, Cloud-native Data Pipelines, Data Storage Solutions, Cost Optimization
- **Programming & Querying:** Python, SQL (T-SQL, PostgreSQL, Spark SQL), PySpark, dbt, Pandas, REST APIs, Bash-, JavaScript/FastAPI, Spark
- **Data Operations & Governance:** Apache Airflow (DAG orchestration), Data Quality Validation & Deduplication, Data Lineage & Governance, Pipeline Monitoring & Incident Triage, CI/CD (GitHub, Azure DevOps), Compliance & Security Best Practices, PII/Privacy Implementation, Data Cataloging, Metadata Management, Data Orchestration, Real-time Data Processing, Streaming Data Processing
- **Reporting & Analytics:** Tableau (dashboard design, ODBC connectivity), Data Modeling for BI, Analytics Dashboard Development, Business Intelligence Integration, Reporting Infrastructure
- **Leadership & Collaboration:** Stakeholder Communication, Business Requirements Translation, Cross-functional Team Leadership, Data Product Development, Technical Documentation, Mentoring & Code Review

PROFESSIONAL EXPERIENCE

Community Dreams Foundation

Feb 2025 - Present

Data Engineer

- Designed and deployed 20 Delta Lake pipelines (PySpark/Airflow) using Airflow for data orchestration across 50+ data sources, achieving 99% on-time SLA, cutting incident resolution time by 30%, and enabling 45 analytics users to access real-time fundraising data 40% faster for a \$50M+ portfolio.
- Developed and rolled out Medallion Architecture (Bronze/Silver/Gold), reducing data onboarding time by 35%, increasing dataset freshness from daily to hourly, and empowering the Finance team to launch 8 more grant analysis projects per quarter while improving donor-retention targeting accuracy by 22% using scalable data storage solutions.
- Consolidated data requirements from 5 business units into 20 reusable products in Azure Data Lake and BigQuery, cutting data request time by 40%, reducing errors by 28%, and implementing metadata management and data cataloging governance policies that sped up reporting for 15 funders.
- Managed daily operations for 50+ data workflows; monitored pipelines, diagnosed failures, executed retries, and tuned performance. Integrated REST APIs and JDBC connectors via CI/CD automation; maintained 99%+ uptime for critical data processes, preventing operational disruptions for downstream analytics teams.

HighRadius

Aug 2021 - Jul 2022

Data Engineer

- Automated Order-to-Cash workflows by building ETL pipelines with AWS Glue, Lambda, and Spark (PySpark) into Redshift across 3 business units, cutting manual processing time by 50% and freeing \$750K+ in annual working capital.
- Optimized 15+ Redshift transformation pipelines (dbt + Git CI/CD), cutting daily query runtime 35% (2.5 hrs → 1.6 hrs). Enabled Finance to close month-end reconciliation 2 days earlier, releasing capital daily and improving cash flow forecasting accuracy by 18%.
- Built collections analytics data products (Tableau/ODBC); improved remittance reconciliation accuracy 20%, reduced manual dispute resolution by 35%, and ensured 98% on-time report delivery to 12 stakeholders. Collections team recovered \$500K+ in flagged accounts using predictive insights from data products.
- Maintained consistent Redshift schemas and Lambda refresh automation; documented data lineage and implemented metadata management practices across 3 business units, cutting data quality errors 25% and reducing support tickets 30%.

School of Engineering and Technology, IUPUI

Mar 2023 - Aug 2023

Teaching Assistant - Data Engineer

- Authored Python and SQL automation scripts leveraging relational databases for grading workflows; reduced feedback turnaround 40% (5 days → 3 days) for 60+ students and improved average assignment scores 15% through standards-based evaluation rubrics and targeted SQL optimization feedback.

- Built SQL-based performance dashboards in Tableau; delivered targeted feedback on security and networking concepts using standardized rubrics. Students' rubric scores increased 18% and demonstrated 22% deeper understanding of SQL optimization principles.

KEY PROJECTS

Full-Stack SQL Learning Platform

SQLit

- Architected and deployed full-stack SQL learning platform with 400+ problems, dual-engine execution (SQLite + PostgreSQL), and real query analysis tooling (EXPLAIN plans, index scans, execution strategies). Reached 500+ users globally with zero-cost barrier to learning SQL-democratizing data literacy.
- Built stateless FastAPI backend with query routing, execution isolation, timeout enforcement, and rate limiting; implemented EXPLAIN plan analysis for query performance insights. Designed relational schemas for user progression analytics, enabling data-driven curriculum improvements.
- Portfolio Impact: Demonstrates full-stack ownership (backend/frontend/database), product thinking (user experience design, engagement), and commitment to mission-driven work (removing barriers to learning). Showcases ability to design scalable systems from zero to 500 users.

ELT Pipeline – Medallion Architecture, Airflow & dbt

Netflix

- Designed and maintained Airflow-based ELT pipelines orchestrating daily batch loads of 20M+ movie ratings from REST APIs into AWS S3 and Snowflake, implementing automated retries, dependency management, and backfill support
- Built Bronze→Silver→Gold Medallion Architecture using PySpark and Spark SQL for data cleaning, null handling, schema enforcement, and dbt CI/CD testing, ensuring 100% data integrity and accelerating query performance by 40%
- Implemented pipeline monitoring, data quality validation, and alerting via Airflow and bash scripts, maintained data lineage and metadata for governance, and achieved zero production incidents over a 6-month period

Real-Time Social Media Analytics – Spark Structured Streaming & Delta Lake

- Designed and implemented real-time data platform using PySpark Structured Streaming and Delta Lake; handled production challenges including late-arriving data handling, shuffle optimization for skewed partitions, and stateful aggregations.
- Orchestrated batch and streaming workflows with Airflow; implemented dependency management, automated retries, manual backfills, and comprehensive pipeline monitoring. Achieved 99.5% uptime and sub-second latency for real-time analytics.
- Applied schema enforcement, versioning, and optimized partitioning strategies within Medallion Architecture. Real-time dashboards reduced decision-making latency 60% for marketing analytics team.

Star Schema & Medallion Architecture

SQL Data Warehouse

- Architected scalable SQL Server data warehouse using Medallion Architecture; integrated ERP and CRM sales data with optimized star schema design for 50+ downstream analytics queries. Warehouse supported 12 concurrent analysts with <2 second query latency.
- Implemented robust version control and documentation in GitHub across 10-member data team; established CI/CD deployment pipeline for schema changes. Boosted deployment efficiency 15% with clear workflows and reduced deployment errors 40%.

EDUCATION

Purdue University

Jan 2023 - Dec 2024

Master of Science, Applied Data Management and Analytics

- **GPA:** 3.7
- **Coursework:** Data Engineering, Advanced SQL, Cloud Data Platforms, Data Governance, Business Intelligence, Data Modeling